MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

FOURTH SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBOT201, MG4DSCBBT200

- PLANT ANATOMY AND REPRODUCTIVE BOTANY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Answer Key

Part A

Short answer

Answer any 10 questions, Each question carries 1 marks

1. **Name the commonly used stain for microscopy. [K] / [CO1]**

Ans: Safranin

2. **What is the use of iris diaphragm in a microscope? [K] / [CO1]**

Ans: To regulate the amount of light entering the microscope

3. **What is a microfibril in the ultrastructure of plant cell wall [U] / [CO2]**

Ans: A microfibril is a fine, crystalline thread-like structure in the plant cell wall composed of bundles of cellulose molecules (micelles), visible under an electron microscope.

4. **Name the substances that form the cementing material between adjacent plant cells.[K] / [CO2]**

Ans: Calcium pectate/ Pectate compounds

5. **What is meant by bidirectional activity of cambium? [K] / [CO3]**

Ans: It is the ability of cambium to produce secondary xylem towards the inner side and secondary phloem towards the outer side

6. **Give one example of anomalous secondary thickening. [K] / [CO3]**

Ans: Anomalous secondary thickening in Bignonia

7. **What is the function of lenticels? [K] / [CO3]**

Ans: Gas exchange

8. **Define the grain pattern in wood. [K] / [CO5]**

Ans: Grain refers to the orientation of wood fibres and other longitudinal elements relative to the longitudinal axis of the stem.

9. **What is male gametophyte? [K] / [CO6]**

Ans: The male gametophyte in flowering plants is the **pollen grain**, which produces male gametes (sperm cells).

10. **What is raphe? [K] / [CO6]**

Ans: Raphe is the ridge formed by the fusion of the funiculus with the ovule body in an anatropous ovule.

11. **What is polyembryony? [K] / [CO6]**

Ans: Polyembryony is the formation of more than one embryo within a single seed.

12. **What is thalamus? [K] / [CO6]**

Ans: Thalamus is the swollen tip of the flower stalk (pedicel) on which floral parts are arranged.

Part B

Short Essay

Answer any 6 questions, Each question carries 5 marks

13. Describe the role of plasmodesmata in plant development. [U] / [CO2]

Ans: Plasmodesmata play a crucial role in plant development by maintaining symplastic continuity between cells, allowing the plant body to function as an integrated system. They facilitate the movement of nutrients, metabolites, and growth regulators between adjacent cells, ensuring coordinated growth and development. They also enable intercellular communication necessary for cell differentiation and tissue formation, and maintain connectivity even in mature and thick-walled cells. **(Any 5 points)**

14. **How can cellulose be converted in to biofuels? [U] / [CO2]**

Ans: Steps in Production of Cellulosic Biofuels

1**. Feedstock Collection**

The raw materials used in cellulosic biofuel production primarily come from agricultural waste and forestry byproducts, which makes them a more sustainable option than food based biofuels.

- Examples: corn stover, wheat straw, rice husk, sawdust
- Energy crops: switchgrass, miscanthus

2. **Conversion of Cellulose to Sugars**

Cellulose can be converted into biofuels using three major techniques

- Enzymatic hydrolysis - Enzymes (cellulase) break cellulose to glucose

- Thermochemical conversion- Uses heat and chemical processes
    - Pyrolysis- : produces bio-oil, syngas, biochar
    - Gasification: produces syngas (H_2, CO, CH_4)
    - HTL: converts biomass to bio-oil using water (good for wet biomass)
- Fermentation: Microorganisms convert sugars to ethanol or butanol
    - Direct Fermentation: This method involves fermenting sugars obtained directly from cellulose. However, due to cellulose's crystalline structure, enzymatic hydrolysis is usually required first to release the sugars.
    - Co-Fermentation: The degradation of cellulose often produces both hexose (glucose) and pentose (xylose) sugars. Co-fermentation uses microorganisms capable of fermenting both types simultaneously, thereby improving total biofuel yield.

3**. Biochemical Upgrading**

After fermentation, biofuels undergo further upgrading processes to enhance their quality and usability. These include distillation, dehydration, and chemical upgrading to refine raw biofuels into usable products like ethanol, butanol, or advanced fuels such as biodiesel.

15. **List out the functions of external secretory tissues [K] / [CO1]**

Ans: Functions of External Secretory Tissues

- Secrete substances (oils, mucilage, terpenes, resins, enzymes, etc.) for various plant activities.
- Attract pollinators by secreting nectar (important for reproduction).
- Provide protection against herbivores by producing irritants, toxins, adhesives, and defensive chemicals.
- Protect plants from pathogens through antimicrobial secretions and resin production.
- Help in wound healing by secretion of resins that seal damaged tissues.
- Aid in water retention and prevent desiccation (mucilage secretion).
- Regulate salt balance and excrete excess salts and water.
- Facilitate ecological interactions (e.g., attracting ants for plant defence via extrafloral nectaries).
- Assist in trapping and digestion of insects in carnivorous plants (enzymes and adhesives).

16. **Briefly explain the characteristics of compression wood. [K] / [CO4]**

Ans: Characteristics of Compression Wood

- Formed as reaction wood on the lower side of the stem due to increased cambial activity.
- Shows eccentric growth rings, making it easily identifiable.
- Heavier (15–40% more) than normal wood of the same species.
- Gradual transition from early wood to late wood (not distinct).
- Tracheids are round in cross-section with intercellular spaces.
- Cell walls thicker (especially spring wood) and higher lignin content.

- Wood is brittle with high and irregular longitudinal shrinkage.
- Lumber twists during drying due to uneven shrinkage.

17. **With suitable examples, explain how anatomical characters used for identification of adulterance on crude drugs. [U] / [CO5] (Give full marks for answers that include four relevant points along with one example).**

Ans: Anatomical characters are essential for identifying adulterants in crude drugs, especially when they are in dried, broken, or powdered forms where external features are lost. Since internal structures remain stable, microscopic examination provides reliable identification.

Each plant species has unique anatomical features such as epidermal cells, stomata, trichomes, fibres, vessels, crystals, and starch grains. These act as diagnostic markers to distinguish genuine drugs from adulterants. Powder microscopy is widely used, as it allows quick analysis of small samples.

Epidermal characters like cell shape and stomatal types (anomocytic, anisocytic, etc.) help detect adulteration. Trichomes are highly specific and resistant, making them reliable indicators. For example, absence of characteristic trichomes suggests substitution.

Supporting tissues such as fibres and sclereids (stone cells) also aid identification. Similarly, xylem vessels show distinct thickening patterns like spiral or pitted, which remain visible after grinding.

Calcium oxalate crystals (raphides, druses, prisms) are species-specific and useful markers. Starch grains and secretory structures like oil cells and ducts further support identification. Microchemical tests (iodine for starch, phloroglucinol for lignin) confirm these features.

Examples:True cinnamon (*Cinnamomum verum*) is distinguished from cassia (*Cinnamomum cassia*) by fibre structure, and senna (*Senna alexandrina*) by its characteristic leaf constants. Haitian bark can be distinguished from Bolivian *Guarea* bark by their differences in phloem fibre arrangement.

18. **How wood is modified for industrial use through physical methods? What are its advantage? [U] / [CO5]**

Ans: Physical modification primarily involves heat treatment, commonly referred to as thermal modification. In this process, wood is exposed to elevated temperatures, generally between 160°C and 260°C, under controlled conditions with limited or no oxygen to prevent combustion. During thermal treatment, hemicelluloses undergo degradation, and the number of free hydroxyl groups decreases. This reduces the wood's ability to absorb moisture. Chemical changes in lignin also occur, leading to improved dimensional stability and resistance to fungal attack. The process will improve qualities like reduced equilibrium moisture content, resistance to fungal decay, enhanced color uniformity (wood becomes darker and aesthetically appealing) and Reduced hygroscopicity. Thermally modified wood is widely used in exterior cladding, decking, window frames, and other applications where dimensional stability is important.

19**. Differentiate between albuminous seeds and exalbuminous seeds? [U] / [CO6]**

Ans: Seeds are classified as albuminous or exalbuminous based on the presence or absence of endosperm in the mature seed.

Albuminous seeds

- Endosperm present in mature seed
- Food stored in endosperm
- Embryo small
- Examples: maize, wheat, castor

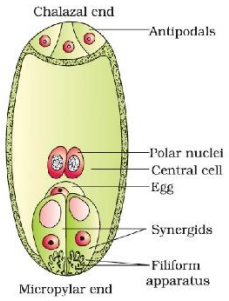
Exalbuminous seeds

- Endosperm absent in mature seed
- Food stored in cotyledons
- Embryo large

20. **Explain the structure of a typical female gametophyte. [U] / [CO6]**

Ans: The female gametophyte generally contains 7 cells with a total of 8 nuclei: one egg cell, two synergid cells, three antipodal cells, and a central cell with two polar nuclei.

- Egg cell: The single cell that fuses with a sperm cell to form the zygote.
- Synergids (2 cells): Located near the micropyle, they help guide the pollen tube for fertilization.
- Antipodal cells (3 cells): Found at the chalazal end, thought to nourish the embryo sac though their function varies.
- Central cell: Contains two polar nuclei which fuse with a sperm nucleus to form the triploid endosperm, supporting the developing embryo.



(Description: 3 marks, Diagram 2 Marks)

Part C

Essay

Answer any 1 questions

Each question carries 10 marks

21. **Explain seasonal activity of cambium and formation of annual rings. Add a note on dendrochronology. [U] / [CO3]**

Ans: The vascular cambium is a lateral meristem responsible for secondary growth in woody plants, and its activity varies with seasonal changes in environmental conditions. During favourable seasons such as spring or the rainy period, the cambium becomes highly active and produces a large amount of secondary xylem and secondary phloem. The xylem elements formed during this period are large in size, thin-walled, and have a wide lumen; this type of wood is known as **early wood (spring wood).** Early wood is generally lighter in colour and less dense. In contrast, during unfavourable seasons such as winter or dry summer, the activity of the cambium slows down or becomes almost dormant. As a result, the xylem elements formed are smaller, thick-walled, and have a narrow lumen; this is called **late wood (autumn wood),** which is darker and denser.

The alternate formation of early wood and late wood during successive seasons leads to the development of distinct concentric rings in the stem, known as annual rings. Each annual ring consists of one band of early wood followed by a band of late wood, representing one year of growth. These rings are clearly visible in the transverse section of the stem due to the difference in structure and density between early and late wood. The number of annual rings helps in determining the age of the plant, and their thickness reflects the growth conditions during each year.

Dendrochronology

Dendrochronology is the scientific method of studying annual rings to determine the age of trees and to analyse past climatic and environmental conditions. By counting the number of annual rings, the exact age of a tree can be estimated. Moreover, the width of the rings provides information about the environmental conditions prevailing during each year. Wider rings indicate favourable conditions such as sufficient rainfall, optimal temperature, and adequate nutrients, while narrower rings suggest unfavourable conditions like drought or environmental stress. Dendrochronology is widely used in fields such as climatology, ecology, and archaeology, where it helps in reconstructing past climates and dating ancient wooden artifacts.

22**. Discuss the process and significance of syngamy and triple fusion [U] / [CO6]**

Ans: In angiosperms, fertilization involves two fusion events—syngamy and triple fusion, collectively known as double fertilization.

Process of Syngamy

- One male gamete (sperm) fuses with the egg cell inside the embryo sac
- Results in the formation of a diploid (2n) zygote
- This process occurs at the micropylar end of the embryo sac

Significance of Syngamy

- Leads to formation of zygote, which develops into the embryo
- Maintains diploid chromosome number

Process of Triple Fusion

- The second male gamete fuses with the two polar nuclei in the central cell
- Forms a triploid (3n) primary endosperm nucleus (PEN)
- This process is called triple fusion
- Occurs in the central cell of the embryo sac

Significance of Triple Fusion

- Develops into endosperm, which provides nourishment to the embryo
- Ensures synchronization of embryo and endosperm development
- Prevents wastage of resources (endosperm forms only after fertilization)